BRITISH INTERNATIONAL

SCHOOL OF SULAYMANIYAH

January

2026

Information & Communication Technology

OVERVIEW

- This chapter covers a wide range of ICT uses in daily life and industry.
- Key areas include communication, modelling, and computer control.
- We will explore financial, medical, and retail applications.
- The role of satellite and recognition systems is examined.
- The content connects with previous knowledge on hardware and software.
- Examples provided are relevant to real-world scenarios.

6.1

Communication

Modern Communication Methods

- ❖ ICT technology drives various communication systems.
- ❖ Common methods include newsletters, posters, and websites.
- ❖ Multimedia presentations and media streaming are widely used.
- ❖ E-publications are replacing traditional paper formats.
- ❖ Mobile communication has revolutionized how we connect.
- ❖ Each method has specific use cases and target audiences.

Production & Use

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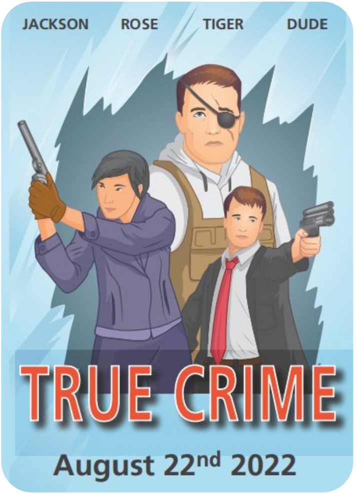
Guidelines for Newsletters
Avoid squeezing too much information onto a single page.
Use clear, easy-to-read fonts like Arial or Trebuchet MS.
Font sizes of 11, 12, or 14 points are recommended.
Use columns to organize text and images effectively.
Avoid excessive capital letters; use bold for headings instead.
Real photos are preferred over clip art for engagement.

Posters for Events

Essential Information

- ❖ Sporting event posters must include the event type and location.
- ❖ Date, time, and admission fees are critical details.
- ❖ Contact details and facility information (e.g., parking) are needed.
- ❖ Movie posters require a representative image of the genre.
- ❖ Release dates and main character lists are standard.
- ❖ Strategic placement allows targeting of specific audiences.



Websites as opposed to print media

- ❖ Websites offer worldwide advertising capability.
- ❖ No need for paper, consumables, or physical distribution.
- ❖ Companies can develop their own or pay for ad space.
- ❖ Requires investment in design, hardware, and maintenance.
- ❖ Security against hackers and pharming is a necessity.
- ❖ Content management systems make self-hosting easier.

Advantages of Websites

- ❖ Sound, video, and animation can be integrated.
- ❖ Hyperlinks allow easy navigation to related pages.
- ❖ Hit counters provide data on visitor numbers.
- ❖ Global reach extends beyond local markets.
- ❖ Updates are easier and faster than reprinting paper.
- ❖ Content cannot be physically defaced or discarded.

Disadvantages of Websites

- ❖ Risk of hacking, modification, or virus introduction.
- ❖ Pharming attacks can redirect users to fake sites.
- ❖ Customers require internet access and a computer/device.
- ❖ Less portable than paper (though smartphones are changing this).
- ❖ Possible for customers to go to undesirable websites
- ❖ Maintenance costs can be high for professional sites.
- ❖ Harder to target a specific local audience compared to posters.
- ❖ The need to find a way for people to find out about the website

Advantages

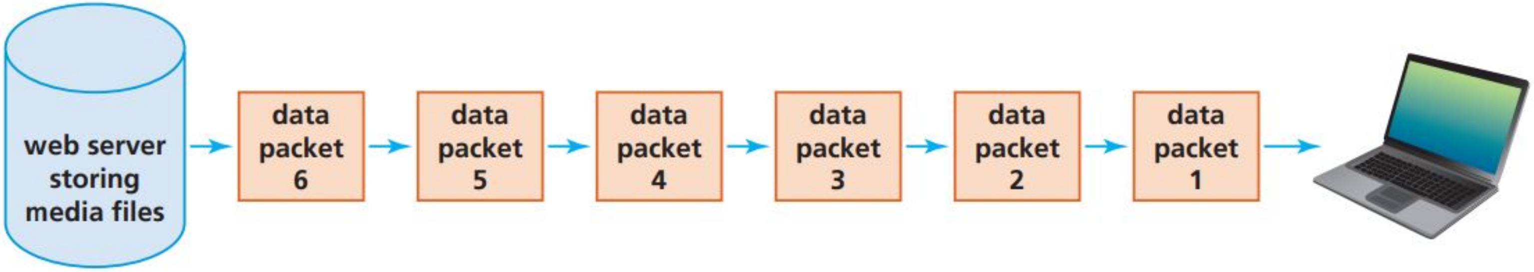
- ❖ Uses animation, video, sound, and music for interest.
- ❖ Produced with software and displayed via projectors.
- ❖ Sound and video grab audience attention effectively.
- ❖ Interactive hyperlinks can access websites or cloud files.
- ❖ Transition effects display facts in chronological order.
- ❖ Presentations can be tailored to different audiences easily.

Disadvantages

- ❖ Specialized equipment can be expensive to procure.
- ❖ Risk of equipment failure during the presentation.
- ❖ Internet access may be required for certain features.
- ❖ Focus can shift to the medium rather than the message.
- ❖ Overuse of effects can result in a "bad" presentation.
- ❖ Too much text or animation can distract the audience.

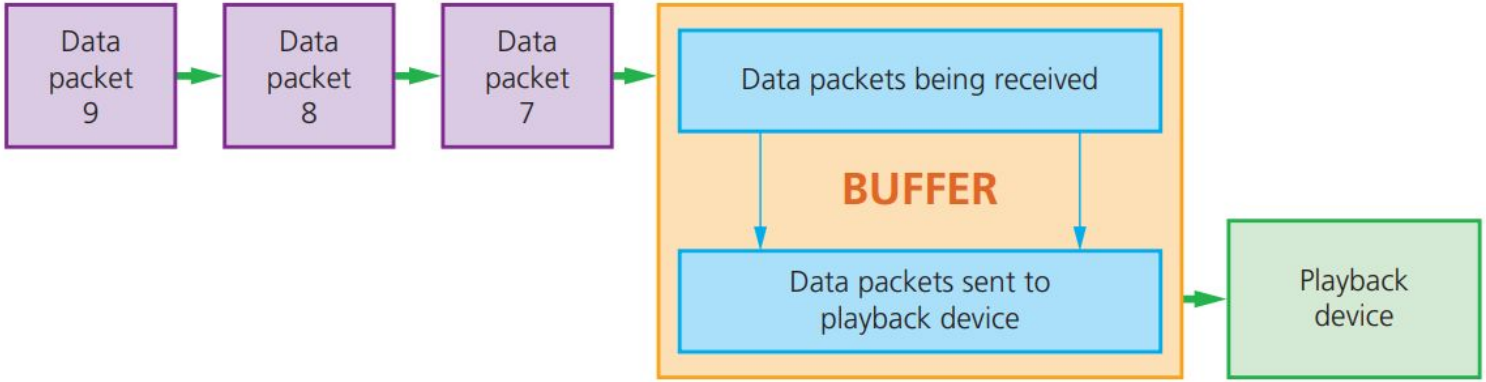
Media Streaming

- ❖ Users watch or listen without downloading the full file.
- ❖ Continuous transmission from a remote server to the device.
- ❖ Data is transmitted and played in real-time.
- ❖ Saves local storage space on HDD or SSD.
- ❖ Avoids long wait times associated with full downloads.
- ❖ Requires a stable internet connection (e.g., 25 Mbps for HD).



How Streaming Works
Buffering and Playback

- ❖ Files are sent as a series of data packets.
- ❖ Data packets are buffered (stored temporarily) in the computer.
- ❖ Buffering ensures smooth playback without freezing.
- ❖ The buffer sends data to the playback device continuously.
- ❖ Large buffers reduce the risk of playback gaps.

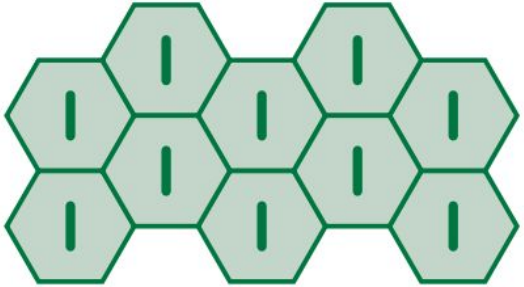


E-publications

- Includes e-books, digital magazines, and newspapers.
- Downloaded to devices for reading via internet connection.
- Navigation is done by swiping the screen.
- Text size can be expanded for better readability.
- Multimedia elements can be embedded in the text.
- Cheaper to produce as there are no printing costs.

Cellular Networks

- ❖ Mobile phones use a network of towers inside cells.
- ❖ Towers allow data transmission across large areas.
- ❖ Signals weaken at cell edges and switch to adjacent towers.
- ❖ Satellite technology links calls between different countries.
- ❖ Devices use SIM cards or wireless internet to connect.
- ❖ Supports SMS, calls, VoIP, and internet access.



Cell showing tower at the centre. Each cell overlaps giving mobile phone coverage

Short Message Service

- ❖ Quick communication via typed text on mobile devices.
- ❖ Recipients do not need to be immediately available.
- ❖ Often free between phones on the same network.
- ❖ Predictive texting speeds up input by completing words.
- ❖ Useful for communicating when phone calls are not possible.
- ❖ Can be sent even if the recipient's phone is off.

Voice Communication

- ❖ Allows staying in touch while on the move.
- ❖ Useful for emergency situations without public phones.
- ❖ Facilitates business and personal calls anywhere.
- ❖ Mobile networks are less stable than landlines.
- ❖ Signal quality depends on network coverage.
- ❖ Essential for keeping contact with co-workers remotely.

Internet Telephony

- ➢ Method for talking to people using the internet.
- ➢ Converts sound into digital data packets for transmission.
- ➢ Accessed via mobile networks or broadband.
- ➢ Calls are often free regardless of global location.
- ➢ Issues include sound quality, echo, and weird sounds.
- ➢ Security risks include identity theft and malware.

Video Calling

- Uses built-in cameras to add video to VoIP calls.
- Cheaper than traditional video conferencing systems.
- Apps like FaceTime and Zoom are common examples.
- Split screens allow seeing multiple people simultaneously.
- Zoom allows recording sessions for later playback.
- Cloud-based services reduce infrastructure costs.

6.2

Modelling Applications

Spreadsheet Models

- Spreadsheets calculate profit and loss based on variables.
- "What-if" scenarios allow testing different financial outcomes.
- Mathematical formulas represent real-world financial situations.
- Used to forecast business performance over time.
- Variables like costs and revenue can be adjusted easily.
- Essential for budgeting and strategic planning.

Civil Engineering Modelling

- Engineers use 3D computer modelling for bridges and buildings.
- Designs are tested long before construction begins.
- Simulations check for structural integrity under stress.
- Tests include wind tunnel effects and earthquake resilience.
- Allows zooming and rotation to view fine details.
- Identifies design flaws early to save money and lives.

Flood Water Management

- Used to predict water levels and potential flood depths.
- Inputs include river cross-sections and bridge dimensions.
- Factors like tides and upstream flows are considered.
- Simulations help plan flood defences and mitigation.
- Past flooding data calibrates the model for accuracy.
- Models update continuously as defences are built.

Venice Example

- Venice uses models to manage flood barriers.
- Simulations predict barrier reactions to flood scenarios.
- Helps determine when to raise barriers based on tides.
- Prevents damage from rising sea levels.
- Complex calculations are handled by computer systems.
- Ensures the safety of the city's infrastructure.

Traffic Modelling

- ❖ Models traffic flow during roadworks or accidents.
- ❖ Tests the effect of different speed limits on flow.
- ❖ Helps design road layouts to minimize congestion.
- ❖ Simulates breakdown scenarios to assess impact.
- ❖ Optimization allows for faster repairs and better flow.
- ❖ Cheaper and safer than physical road closures for testing.

Traffic Light Simulation

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